

Vedanth Prasanna Bharadwaj

(623) 210-8716 | vedanthpb1@gmail.com | linkedin.com/in/vedanth-p-bharadwaj | github.com/vedanthpbharadwaj | vedanthpbharadwaj.github.io

PROFESSIONAL SUMMARY

Backend & cloud-focused software engineer with M.S. in Data Science & Engineering (ASU, GPA: 3.9). Production experience building serverless APIs, database gateways, and cost/latency-optimized systems on AWS. Hands-on RAG pipeline engineering (LangChain, FAISS, ChromaDB) and AI/ML modeling across multiple domains. Seeking full-time SWE/SDE roles in backend, cloud, or AI/data engineering.

F1-OPT | STEM OPT extension eligible

EDUCATION

Arizona State University

Master of Science, Data Science, Analytics & Engineering – GPA: 3.9/4.0

Tempe, AZ

Jan 2024 – Dec 2025

B.N.M. Institute of Technology, Visvesvaraya Technological University

Bachelor of Engineering, Information Science & Engineering

Bengaluru, India

Aug 2018 – Jul 2022

TECHNICAL SKILLS

Languages: Python, Java, SQL (MySQL, PostgreSQL), JavaScript, R, C++

Frameworks & Libraries: Spring Boot, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, Angular

Cloud & Backend: AWS (Lambda, API Gateway, S3, Aurora PostgreSQL, DynamoDB, Step Functions, CloudWatch, AppConfig),

Serverless Architecture, Microservices, REST API Design, ETL Pipelines, Cost Optimization

AI/ML: XGBoost, LSTM, Bi-LSTM, CNN (VGG16), Random Forest, Time-Series Forecasting, Feature Engineering, Anomaly Detection

LLM & RAG: LangChain, FAISS, ChromaDB, RAG, sentence-transformers, Groq LLM, Prompt Engineering, Claude Code

Analytics & Tools: Power BI, Tableau, Git, GitHub, OOP & Design Patterns, Data Structures & Algorithms

EXPERIENCE

AWS Cloud Developer, MindSpark, Mesa, AZ

Feb 2026 – Present

- Built and maintained the core database gateway (RDS-Processing Lambda) for the Kairos platform - dynamic SQL, type-safe RDS Data API parameters, referential integrity across **21 tables**; added per-caller metadata layer for Aurora cycle tracing and cost attribution
- Diagnosed a **4x Aurora Serverless ACU cost spike** (\$3→\$12/day) via CloudWatch Database Insights - traced to a background Lambda polling Aurora 24/7 with unfiltered SELECT * and a batch system generating sustained write load; coordinated evidence-backed fixes across **four engineering teams** and authored a **5-chapter engineering certification course** with graded assessments and runbooks to institutionalize findings
- Resolved a **25–30s production latency spike**: traced to Aurora auto-pause delays and blocking AppConfig initialization via CloudWatch; refactored to lazy-init caching and eliminated dead SQL joins, cutting response time to **150–200ms (~95% reduction)**
- Designed multi-table transactions with automatic rollback, partial-update safety, and stepped retry backoff for Aurora cold starts; implemented client-side UUID pre-generation for FK dependency chains achieving **zero data-integrity violations** in production; managed staged deployment (dev → staging → prod) via Lambda versioning

Software Engineer Trainee, Cognizant, Bengaluru, India

Nov 2022 – Jul 2023

- Completed 8-month enterprise Java Full Stack training program (Spring Boot, RESTful APIs, microservices, AWS); maintained **95% average** across 20+ assessments in SQL, OOP design patterns, and cloud architecture
- Built 3 capstone projects simulating enterprise healthcare systems — microservices decomposition, relational schema design, REST API contracts, and end-to-end integration testing

PROJECTS

RAG Document QA Pipeline (Personal) | Python, LangChain, FAISS, ChromaDB, sentence-transformers, Groq LLM

May 2026

- Built end-to-end RAG pipeline: PDF ingestion (PyMuPDF), recursive text chunking, sentence-transformer embeddings, dual vector store (FAISS + ChromaDB), confidence-filtered retrieval, and LLM-powered answer generation with source attribution
- Implemented custom retriever with score thresholding, top-k filtering, and context window management across multi-document corpora using LangChain orchestration

Urban Mobility Data Pipeline | Python, AWS Lambda, S3, DynamoDB, XGBoost, Power BI, SQL

Aug – Dec 2025

- Led 5-person team building serverless AWS pipeline processing **41M+** NYC taxi records; automated data validation and anomaly detection, improving data quality to **96%** and reducing manual effort by **80%**
- Engineered **58+** temporal, spatial, and weather features; trained XGBoost model achieving **$R^2 = 0.90$ (30% improvement)** over baseline; Power BI dashboard tracking **12+ KPIs** across 265 city zones

Energy Consumption Forecasting | Python, TensorFlow, LSTM, Random Forest, Scikit-learn, Tableau

Jan – May 2024

- Led 4-person team building energy demand forecasting pipeline on **1M+** hourly records; engineered 15+ temporal and environmental features; trained LSTM and Random Forest models achieving **97.8% accuracy (18% improvement)** over baseline)
- Built end-to-end ML pipeline covering preprocessing, sliding-window feature generation, cross-validation, and anomaly detection; Tableau dashboards projected **12% cost savings (~\$50K/year)**

Indian Sign Language Recognition - Published | Python, OpenCV, MediaPipe, LSTM, TensorFlow

Oct 2021 – Jun 2022

- Built real-time ISL interpreter using hand landmark extraction and LSTM sequence modeling; achieved **97% accuracy** across varied lighting and backgrounds; published in *IJRTI*, Vol. 7, Issue 6, 2022

CERTIFICATIONS

Anthropic: AI Fluency - Framework & Foundations

May 2026

Google Data Analytics Professional Certificate (SQL, Tableau, R, Data Cleaning)

Jun 2025

Microsoft: Data Analysis and Visualization with Power BI

Nov 2025